Chernick's Prime Triple: Exact Local Obstructions and a Carmichael Consequence

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Abstract

OEIS A033502 records numbers $(6m+1)(12m+1)(18m+1)$ whose three linear factors are prime. We give a self-contained verification of the Carmichael property, then count *exactly* the residue classes that avoid every prime obstruction up to a fixed bound. The resulting positive local Euler product explains why this prime triple is admissible. These are established, elementary consequences of Chernick's construction and local sieving; no infinitude theorem or novelty claim is made.

Sequence and question

For integers $m \geq 1$, put

$$L_1(m) = 6m+1, \quad L_2(m) = 12m+1, \quad L_3(m) = 18m+1, \qquad C(m) = \prod_{j=1}^{3} L_j(m).$$

OEIS A033502 consists of the values $C(m)$ for which all $L_j(m)$ are prime; the corresponding parameters are in A046025. Its first terms are 1729, 294409, and 56052361, from $m = 1, 6, 35$. As recorded in OEIS, Dickson's conjecture implies infinitude of this family; that infinitude is not established unconditionally. We instead settle the concrete finite-congruence question: how many residue classes can survive all obstructions from primes at most y?

Theorem 1 (Chernick verification and exact local sieve)**.** *For $m \geq 1$ and any real $y \geq 3$, the following statements hold.*

1. *If $L_1(m), L_2(m), L_3(m)$ are all prime, then $C(m)$ is a Carmichael number. Moreover,*

$$C(m) - 1 = 36m(1 + 11m + 36m^2), \qquad C(m) \equiv 1 \pmod{72}.$$

2. *Let $\nu(\ell)$ denote the number of classes $m \pmod{\ell}$ for which $\ell \mid L_1(m)L_2(m)L_3(m)$, where ℓ is prime. Then*

$$\nu(2) = \nu(3) = 0, \qquad \nu(\ell) = 3 \quad (\ell \geq 5).$$

3. *Let $P(y) = \prod_{\ell \leq y,\, \ell \text{ prime}} \ell$. The number of classes $m \pmod{P(y)}$ satisfying $\gcd(C(m), P(y)) = 1$ is exactly*

$$\boxed{6 \prod_{\substack{5 \leq \ell \leq y \\ \ell \text{ prime}}} (\ell - 3)}.$$

In particular this number is positive for every y, so infinitely many positive integers m evade any fixed finite *collection of prime-divisor obstructions.*

Proof. The three prime factors in (1) are distinct, so $C(m)$ is squarefree and composite. Direct multiplication gives the displayed identity, whence each of $L_j(m) - 1 \in \{6m, 12m, 18m\}$ divides $C(m) - 1$. If a is coprime to $C(m)$, Fermat's little theorem yields $a^{C(m)-1} \equiv 1$ modulo each $L_j(m)$, hence modulo their product by the Chinese remainder theorem. Thus $C(m)$ is Carmichael. Also $m(1 + 11m + 36m^2)$ is even: this is immediate when m is even, and the parenthesis is even when m is odd. Hence $72 \mid C(m) - 1$.

For (2), each coefficient $6, 12, 18$ vanishes modulo 2 or 3, leaving $L_j(m) \equiv 1$. For a prime $\ell \geq 5$, each coefficient is invertible and the three roots

$$m \equiv -6^{-1}, \quad -12^{-1}, \quad -18^{-1} \pmod{\ell}$$

are distinct: equality of two roots would force ℓ to divide 6 or 12, impossible for $\ell \geq 5$. Exactly $\ell - 3$ classes therefore avoid divisibility by ℓ.

For (3), the Chinese remainder theorem independently combines the 2 available classes modulo 2, the 3 modulo 3, and the $\ell - 3$ surviving classes for each prime $\ell \geq 5$. This gives the exact product. Since every factor is positive, at least one full residue class modulo $P(y)$ survives; each such class has infinitely many positive representatives. □

A positive local constant, but no infinitude proof

The standard three-linear-form local correction factor is

$$\mathfrak{S} = \prod_{\ell \text{ prime}} \frac{1 - \nu(\ell)/\ell}{(1 - 1/\ell)^3} = 27 \prod_{\substack{\ell \geq 5 \\ \ell \text{ prime}}} \frac{1 - 3/\ell}{(1 - 1/\ell)^3}.$$

Every factor is positive, and the factors for $\ell \geq 5$ equal $1 + O(\ell^{-2})$; since $\sum_\ell \ell^{-2} < \infty$, the product converges to a strictly positive, finite constant. This is *local admissibility*, not a proof that all three forms are simultaneously prime infinitely often. For example, for $y = 7$ there are exactly $6(5 - 3)(7 - 3) = 48$ admissible classes modulo 210.

Scope and provenance. The Carmichael implication is Chernick's 1939 construction, and the residue count follows by elementary modular arithmetic and the Chinese remainder theorem. The displayed constant is the conventional local factor for three linear forms. No originality, solution of Dickson's conjecture, or independent human review is claimed. For $m > y/6$, a prime triple must be in one of the surviving classes, but survival by itself never establishes primality.

References

[1] OEIS Foundation, A033502, Chernick's Carmichael numbers; see also A046025 (parameters), accessed November 2026.

[2] J. Chernick, "On Fermat's simple theorem," *Bulletin of the American Mathematical Society* **45** (1939), 269–274. doi:10.1090/S0002-9904-1939-06953-X.

[3] L. E. Dickson, "A new extension of Dirichlet's theorem on prime numbers," *Messenger of Mathematics* **33** (1904), 155–161. (The prime-tuple conjecture is not assumed in the theorem above.)